Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Preprocess and GAF Transform
2. Abbreviated Title	Preprocess GAF
3. Use Case Id	UC-10
4. Actors	ML Engine (Supporting)
5. Description	The ML Engine preprocesses raw patient data (normalization, encoding, missing value handling) and applies the Gramian Angular Field (GAF) transformation to convert tabular features into 2D image representations for the CNN branch.
5.1. Pre Conditions	1. Raw patient feature vector (13 features) is received from UC-01. 2. Preprocessing pipeline configuration is loaded.
5.2. Task Sequence	1. System receives the raw 13-feature vector. 2. System applies min-max normalization scaled to [-1, 1]. 3. System encodes categorical features (Gender, Chest Pain Type, etc.). 4. System applies Gramian Angular Field transformation. 5. System outputs the normalized vector and GAF image matrix.
5.3. Post Conditions	1. Normalized feature vector is ready for the LSTM branch. 2. GAF image matrix is ready for the CNN branch.
6. Modification History	Date: September 11, 2026
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